

Finite-real stability products for generalized Boole maps: divisor resonances and a compensated parabolic limit

Anonymous Authors

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Abstract

We determine the stability product formed from all finite real periodic points of the generalized Boole map $T(x) = ax - b/x$, for every $a, b > 0$ in the original iteration clock. The physical periodic set excludes infinity, so its product differs from the classical projective-circle determinant even when the omitted orbit has measure zero. A rational fixed-point index identity gives the exact weighted sums in the subcritical, parabolic and supercritical regimes. We use them to classify every zero and pole of the subcritical product and prove that it is entire precisely at the parameters $a = (1 - 2a)^{2m}$, $m \geq 1$, in $(0, 1/2)$. At $a = 1$ the normalized germ is $(1 - u)^{1/2} \exp(\text{Li}_2(u)/2)$, where Li_2 is the dilogarithm, and has no meromorphic continuation through $u = 1$. The subcritical germs converge to it locally uniformly on $|u| < 1$; the supercritical germs have the same limit after deleting the complete stability factors of the two newborn real fixed orbits, whereas the unreduced family does not. Uniform bounds on all iterate weights justify both limits. The classical phase geometry is included to fix the periodic set and its multipliers. The resulting products do not identify a natural transfer operator on the punctured real domain or a target arithmetic spectrum.

1 The finite-real question

A periodic-orbit product can change when one orbit is removed, even if the underlying measure space is unchanged modulo null sets. Generalized Boole maps provide an explicit setting in which this distinction persists through a parabolic transition. For $a, b > 0$ set

$$T_{a,b}(x) = ax - \frac{b}{x}, \quad X_{a,b} = \mathbb{R} \setminus \bigcup_{j \geq 0} T_{a,b}^{-j}\{0\}. \quad (1.1)$$

Inverse images in this definition are computed for the rational map on the sphere, and only finite real points are deleted. We study its restriction to $X_{a,b}$, with one unit of time equal to one original iteration. Infinity is not a physical state. Write $q = 2a - 1$ and define

$$\tau_n(a) = \sum_{x \in \text{Fix}_{X_{a,b}} T_{a,b}^n} \frac{1}{(T_{a,b}^n)'(x) - 1}, \quad D_a(u) = \exp \left(- \sum_{n \geq 1} \frac{\tau_n(a)}{n} u^n \right). \quad (1.2)$$

We prove below that every denominator is positive and that the series converges on $|u| < 1$. This notation defines a stability germ, not an operator determinant by assumption. The scale b disappears from the formulas because $x = \sqrt{b}y$ conjugates $T_{a,b}$ to $T_{a,1}$ without changing time.

Theorem 1.1 (Stability weights and continuation). *For all $a, b > 0$ and every $n \geq 1$, every finite real fixed point of $T_{a,b}^n$ is simple and has multiplier greater than one. Its weighted sum is*

$$\tau_n(a) = \begin{cases} \frac{2}{\frac{1-q^n}{n-1} - \frac{1}{1-a^n}}, & 0 < a < 1, \\ \frac{2n}{1}, & a = 1, \\ \frac{1}{a^n - 1}, & a > 1. \end{cases} \quad (1.3)$$

The normalized germ has the following continuations:

$$D_a(u) = \begin{cases} (1-u) \frac{\prod_{j \geq 1} (1 - q^j u)^2}{\prod_{k \geq 1} (1 - a^k u)}, & 0 < a < 1, \\ (1-u)^{1/2} \exp(\text{Li}_2(u)/2), & a = 1, \\ \prod_{k \geq 1} (1 - a^{-k} u), & a > 1. \end{cases} \quad (1.4)$$

For $0 < a < 1$, the product is entire if and only if

$$0 < a < \frac{1}{2}, \quad a = (1 - 2a)^{2m} \quad \text{for some } m \in \mathbb{N}. \quad (1.5)$$

For each $m \geq 1$ there is exactly one such parameter; the first is $a = 1/4$. All remaining zero and pole multiplicities are classified in Proposition 4.1.

Theorem 1.2 (Compensated parabolic limit). *The critical germ D_1 is not meromorphic through $u = 1$. For $a > 1$ define*

$$F_a(u) = \prod_{j \geq 1} (1 - q^{-j} u)^2, \quad D_a^{\text{red}}(u) = \frac{D_a(u)}{F_a(u)}. \quad (1.6)$$

The factor F_a is exactly the complete primitive stability factor of the two finite real fixed orbits, including all repetitions. On $|u| < 1$,

$$D_a \longrightarrow D_1 \quad (a \uparrow 1), \quad D_a^{\text{red}} \longrightarrow D_1 \quad (a \downarrow 1) \quad (1.7)$$

locally uniformly. The unreduced supercritical germs do not have this limit. No statement of uniform convergence includes the boundary point $u = 1$.

Classical inputs and the scope of the increment. The statistical phase picture is established prior work: Umeno and Okubo describe the Cauchy, infinite-measure and dissipative regimes and compute Lyapunov exponents [3]. Mendoza Mendoza and Ruiz Leal study the more general map $ax - b/x + c$ [2]. Their deleted-prepare domain (Section 3) and symbolic descriptions are directly relevant here. In particular, their Theorems 4.2 and 4.4 for the critical and $1/2 < a < 1$ regimes already yield the corresponding unweighted periodic counts; their Theorem 4.5 gives the supercritical binary Cantor dynamics. We do not claim the domain, phase structure, symbolic coding or those counting consequences as new results.

The closest operator input is the finite-Blaschke spectrum of Bandtlow, Just and Slipantschuk [1]. It concerns natural holomorphic function spaces on the full circle. In the subcritical regime the Cayley map conjugates $T_{a,b}$ to a Blaschke product, but the physical periodic set omits the circle

point corresponding to infinity. Section 6 gives its exact missing factor. This deletion must not be inferred from an L^p quotient: removing a null orbit alone does not remove a trace contribution.

The single question addressed here is the complete *weighted* finite-real product across the transition: the exact divisor, its exceptional entire parameters, and the two-sided limit with the dynamically specified supercritical compensation. We derive these statements without invoking a natural transfer-operator realization on $X_{a,b}$. The literature comparison is bounded and does not establish publication priority. Sections 2–3 identify the physical cycles and compute their indices; Sections 4–5 prove the continuation and limiting statements.

2 The physical domain and its periodic points

We include the phase geometry to justify the periodic set and multipliers used in (1.2), not as an independent contribution.

2.1 Inverse branches and invariant measures

The deleted set in (1.1) is countable, since a degree-two rational map has at most 2^j inverse images of a specified point at depth j . Every finite periodic point avoids zero: zero maps to infinity, which is fixed. Thus every finite real periodic point belongs to $X_{a,b}$. Direct calculation gives

$$T'(x) = a + \frac{b}{x^2} > 0, \quad h_{\pm}(y) = \frac{y \pm \sqrt{y^2 + 4ab}}{2a}, \quad h'_{\pm}(y) = \frac{1}{2a} \left(1 \pm \frac{y}{\sqrt{y^2 + 4ab}} \right) > 0. \quad (2.1)$$

Each half-line maps increasingly onto the real line. Since $h'_+ + h'_- = 1/a$, change of variables proves, for every measurable A ,

$$m(T^{-1}A) = a^{-1}m(A), \quad (2.2)$$

where infinite values are allowed and the countable deletions do not change Lebesgue measure. At $a = 1$ this proves invariance of infinite Lebesgue measure; it is not an ergodicity argument.

Suppose $0 < a < 1$, and set $c = \sqrt{b/(1-a)}$. The Cayley map $C(z) = (z - ic)/(z + ic)$ sends the upper half-plane to the disk and infinity to 1. Substitution gives

$$CTC^{-1}(w) = B_q(w) = w \frac{w + q}{1 + qw}, \quad -1 < q < 1. \quad (2.3)$$

For $|w| < 1$, $|1 + qw|^2 - |w + q|^2 = (1 - q^2)(1 - |w|^2) > 0$. Therefore $|B_q(w)| < |w|$ for nonzero disk points, so none is periodic. Reflection in the circle gives the analogous assertion outside the closed disk. The only off-circle periodic points are 0 and infinity, both fixed with local multiplier q ; in the original coordinate these are ic and $-ic$.

For $B_q(e^{i\theta}) = e^{i\psi(\theta)}$, logarithmic differentiation yields

$$\psi'(\theta) = 1 + \frac{1 - q^2}{|e^{i\theta} + q|^2} \geq 1 + \frac{1 - |q|}{1 + |q|} > 1. \quad (2.4)$$

All circle cycles are simple and repelling. At a finite real periodic point, the coordinate derivatives in the full-period conjugacy cancel, so the real multiplier equals the positive angular multiplier.

For completeness, B_q preserves normalized angular measure: the average of $B_q(w)^k$ on the circle is zero for every positive integer k , by the holomorphic mean-value formula and $B_q(0) = 0$. The

negative Fourier modes follow by conjugation; trigonometric approximation gives invariance for all continuous functions. Pullback under C yields

$$d\mu_{a,b}(x) = \frac{c}{\pi(x^2 + c^2)} dx. \quad (2.5)$$

The deleted set is null for this probability. No uniqueness or mixing claim is needed.

2.2 Supercritical survivor and exact escape

Let $a > 1$, $r = \sqrt{b/(a-1)}$ and $I = [-r, r]$. Its endpoints are fixed with multiplier $q = 2a - 1 > 1$, and the inverse images are disjoint:

$$h_-(I) = [-r, -(a-1)r/a], \quad h_+(I) = [(a-1)r/a, r]. \quad (2.6)$$

Each inverse has derivative less than $1/a$ there. The cylinders $h_{\epsilon_1} \cdots h_{\epsilon_n}(I)$ have diameter at most $2ra^{-n}$; distinct words give disjoint cylinders by injectivity and the first-level gap. Nested cylinders therefore code a unique point for every infinite sign sequence. This is a homeomorphism from the one-sided binary shift space onto

$$K = \bigcap_{n \geq 0} T^{-n}I, \quad (2.7)$$

and T deletes the first sign. No point of K meets zero, so $K \subset X_{a,b}$.

For $x > r$, $T(x) - x = (a-1)x - b/x > 0$. A forward orbit starting there is increasing and cannot have a finite limit: such a limit would be a fixed point larger than r . Hence it tends to positive infinity. Oddness gives negative escape for $x < -r$. Every point of $X_{a,b} \setminus K$ eventually leaves I and escapes, so all finite real cycles lie in K . Iterating (2.2) gives the exact initial-Lebesgue survival law

$$m(T^{-n}I) = 2ra^{-n}, \quad m(K) = 0, \quad -\lim_{n \rightarrow \infty} \frac{1}{n} \log \frac{m(T^{-n}I)}{m(I)} = \log a. \quad (2.8)$$

This is not an assertion of a Lebesgue invariant probability on the repeller.

2.3 All iterates and the parabolic multiplicity

For nonreal $z = x + iy$,

$$\Im T(z) = y \left(a + \frac{b}{|z|^2} \right). \quad (2.9)$$

If $a \geq 1$, the absolute imaginary part strictly increases at every finite step. A nonreal point cannot reach a real pole and cannot be periodic. Every finite real periodic multiplier is greater than one, since each derivative is greater than one. Together with (2.4) this proves positive simple repulsion in all three regimes.

The rational degree of T^n is 2^n . Its homogeneous fixed-point equation is a nonzero homogeneous polynomial of degree $2^n + 1$, so it has $2^n + 1$ projective roots counted with local multiplicity. To account for infinity, use $w = 1/z$:

$$g_a(w) = \frac{1}{T(1/w)} = \frac{w}{a - bw^2}. \quad (2.10)$$

If $a \neq 1$, the multiplier of g_a^n at zero is $a^{-n} \neq 1$, giving multiplicity one. If $a = 1$, write $g_1^n(w) = w + A_n w^3 + B_n w^5 + O(w^7)$. Composition with $g_1(w) = w + bw^3 + b^2 w^5 + O(w^7)$ yields $A_{n+1} = A_n + b$ and $B_{n+1} = B_n + 3bA_n + b^2$. Thus

$$g_1^n(w) = w + nbw^3 + \frac{n(3n-1)}{2}b^2w^5 + O(w^7). \quad (2.11)$$

The nonzero cubic coefficient makes infinity a fixed point of multiplicity three for every n , not a simple fixed point of multiplier one.

In the subcritical case exactly two off-real fixed points, each simple for every iterate, were found in (2.3). In the other regimes there are none by (2.9). Subtracting these and the correct infinity multiplicity proves

$$N_n := \# \text{Fix}_{X_{a,b}} T^n = \begin{cases} 2^n - 2, & 0 < a \leq 1, \\ 2^n, & a > 1. \end{cases} \quad (2.12)$$

The primitive orbit counts follow by Möbius inversion: $P_n = n^{-1} \sum_{d|n} \mu(n/d) N_d$. Consequently the unweighted Artin–Mazur zeta is $(1-u)^2/(1-2u)$ for $a \leq 1$ and $1/(1-2u)$ for $a > 1$, initially on $|u| < 1/2$. These elementary counts do not determine the weighted product.

3 A residue calculation of every stability weight

The non-simple infinity index, rather than its multiplier or fixed multiplicity, controls the critical weighted sum. Let F be a rational map fixing infinity and consider $\eta = dz/(z - F(z))$. A finite pole of F is a zero of this form. Its other possible finite poles are fixed points, and at a simple fixed point its residue is $1/(1 - F'(z))$. For $g(w) = 1/F(1/w)$ a direct coordinate transformation gives

$$\eta = \left(\frac{1}{w - g(w)} - \frac{1}{w} \right) dw. \quad (3.1)$$

Let $I_\infty(F)$ be the coefficient of w^{-1} in $1/(w - g(w))$. The residue theorem on the sphere gives

$$\sum_{\substack{F(z)=z \\ z \text{ finite}}} \text{Res}_z \frac{d\zeta}{\zeta - F(\zeta)} + I_\infty(F) = 1. \quad (3.2)$$

The subtractive $-1/w$ in (3.1) is necessary; it is the source of the constant on the right.

Apply this to $F = T^n$. All finite fixed points are simple by Section 2, including the subcritical off-real pair. For $a \neq 1$, the index is $I_\infty(T^n) = 1/(1 - a^{-n})$. At $a = 1$, (2.11) instead gives

$$\frac{1}{w - g_1^n(w)} = -\frac{1}{nbw^3} + \frac{3n-1}{2n} \frac{1}{w} + O(w), \quad I_\infty(T^n) = \frac{3n-1}{2n}. \quad (3.3)$$

In particular, the fixed multiplicity 3 is not this index.

Changing signs in the finite terms of (3.2) yields

$$\sum_{\substack{T^n(z)=z \\ z \text{ finite}}} \frac{1}{(T^n)'(z) - 1} = I_\infty(T^n) - 1. \quad (3.4)$$

Away from criticality the right side is $1/(a^n - 1)$. For $a < 1$, the two nonreal fixed points have multiplier q^n and each contributes $1/(q^n - 1)$, so they must be subtracted. For $a \geq 1$ all finite fixed points are real. Substitution proves (1.3), including the critical value $(n-1)/(2n)$.

All weights are nonnegative, as also follows from their original positive denominators. For $a \leq 1$, $N_1 = \tau_1 = 0$. At criticality the two-cycle $\{\sqrt{b/2}, -\sqrt{b/2}\}$ has multiplier 9, so its two points contribute $2/(9-1) = 1/4$, consistent with the all-iterate formula at $n = 2$. This example is an exact consequence, not a substitute for the index argument.

4 Primitive products and the complete divisor

For fixed $0 < a < 1$, the denominators in (1.3) are uniformly separated from zero as n varies: $1 - a^n \geq 1 - a$ and $1 - q^n \geq 1 - |q|$. Thus τ_n is bounded. At criticality it is bounded by $1/2$, and above criticality it decays exponentially. The logarithmic series in (1.2) therefore converges absolutely on $|u| < 1$.

For a finite-real primitive orbit γ with least period p_γ , let $\Lambda_\gamma = (T^{p_\gamma})'(x) > 1$ at any point on it. Its s -fold repetition has multiplier Λ_γ^s , and

$$\frac{1}{\Lambda_\gamma^s - 1} = \sum_{j \geq 1} \Lambda_\gamma^{-js}. \quad (4.1)$$

The sum obtained after replacing u by $|u|$ is finite. Grouping first by primitive orbits and then by j is therefore justified and gives

$$D_a(u) = \prod_{\gamma \text{ primitive}} \prod_{j \geq 1} \left(1 - \frac{u^{p_\gamma}}{\Lambda_\gamma^j} \right), \quad |u| < 1. \quad (4.2)$$

This includes only finite-real primitive orbits. We do not assert global convergence of the primitive product.

Expand the weights (1.3) into geometric series and use $-\sum_{n \geq 1} v^n/n = \log(1 - v)$ on the disk. Absolute convergence permits interchanging the sums and proves (1.4). Here $\text{Li}_2(u) = \sum_{n \geq 1} u^n/n^2$ and the square-root branch is normalized to one at zero. Each separate product with geometric bases q^j , a^k , or a^{-k} converges locally uniformly on $\mathbb{C}$, since the bases are absolutely summable. Away from the listed zeros its tail is nonzero by absolute convergence of the tail logarithms. Hence the subcritical quotient is meromorphic on $\mathbb{C}$, while the supercritical product is entire with simple zeros $u = a^k$, $k \geq 1$.

Proposition 4.1 (All subcritical cancellations). *Let $0 < a < 1$. The point $u = 1$ is always a simple zero. If $q \neq 0$, the numerator has double zeros at q^{-j} , $j \geq 1$; the denominator has simple zeros at a^{-k} , $k \geq 1$. A denominator zero not coinciding with a numerator zero is a simple pole. A coincidence is a simple zero, and every remaining q^{-j} is a double zero. The coincidence equation is exactly $a^k = q^j$.*

If $q = 0$ there are no coincidences. If $q > 0$ and $\log a / \log q = M/N$ in lowest positive terms, coincidences are exactly $k = N\ell$, $j = M\ell$, $\ell \geq 1$; for an irrational ratio there are none. If $q < 0$ and $\log a / \log(q^2) = M/N$ in lowest positive terms, coincidences are exactly $k = N\ell$, $j = 2M\ell$; for an irrational ratio there are none.

Proof. All zeros within each geometric sequence are distinct, and none equals one. Thus the orders follow directly from (1.4), with no hidden zeros in the product tails. For $q > 0$ the equation is equivalent to $k \log a = j \log q$. For $q < 0$ it first forces j even and then reduces to the same rational commensurability statement for q^2 . This proves the full classification. $\square$

To finish the entire classification in Theorem 1.1, first take $q \geq 0$. If $q = 0$, no denominator point cancels. If $q > 0$, then $q < a < 1$, so no q^j equals a : the first denominator zero cannot cancel. If $q < 0$, entireness likewise requires cancellation of that first point, which means $a = q^{2m} = (1 - 2a)^{2m}$ for some $m \geq 1$. Conversely this equality makes every denominator factor equal to the numerator factor with index $2mk$, so all poles cancel. For each fixed m , the function $(1 - 2a)^{2m} - a$ is continuous and strictly decreasing on $(0, 1/2)$, with endpoint limits 1 and $-1/2$. It has exactly one root a_m . The roots are distinct, since two distinct powers of a number in $(0, 1)$ cannot agree, and $a_1 = 1/4$.

At such a root the surviving entire product is

$$D_{a_m}(u) = (1 - u) \prod_{j \geq 1} (1 - q^j u)^{2^{-1}\{2m|j\}}. \quad (4.3)$$

An abstract self-adjoint trace-class operator realizes this product: take a real diagonal operator with entry 1 once and q^j with the displayed multiplicity. Its absolute trace sum is at most $1 + 2|q|/(1 - |q|)$. Similarly, the diagonal entries a^{-k} realize the supercritical entire product. These are constructions from the already known factors, not identifications with a natural dynamical transfer operator.

5 The critical singularity and both limiting directions

The critical formula (1.4) yields

$$\frac{D'_1(u)}{D_1(u)} = -\frac{1}{2(1-u)} - \frac{\log(1-u)}{2u}, \quad |u| < 1. \quad (5.1)$$

Consequently

$$\lim_{u \uparrow 1} (u-1) \frac{D'_1(u)}{D_1(u)} = \frac{1}{2}. \quad (5.2)$$

A nonzero meromorphic function near one has the form $(u-1)^k h(u)$, with $k \in \mathbb{Z}$ and $h(1) \neq 0$ holomorphic, and the same limit would be the integer k . This proves the non-meromorphic assertion. It excludes an entire ordinary trace-class Fredholm determinant with this germ. It does not give a natural boundary on the whole unit circle: the logarithm and dilogarithm admit compatible local continuations near its other points.

For fixed n , binomial expansion gives

$$\frac{1}{1 - (1-h)^n} = \frac{1}{nh} + \frac{n-1}{2n} + O_n(h), \quad (5.3)$$

$$\frac{1}{(1+h)^n - 1} = \frac{1}{nh} - \frac{n-1}{2n} + O_n(h). \quad (5.4)$$

The first formula with $a = 1 - \varepsilon$, $q = 1 - 2\varepsilon$ shows $\tau_n(a) \rightarrow (n-1)/(2n)$ as $a \uparrow 1$. A uniform bound is still needed to pass through the full logarithmic series. For $1/2 < a < 1$, the relation $q \leq a^2$ follows from $a^2 - q = (a-1)^2$. Convexity of x^n on the positive line, with $a = (1+q)/2$, gives $1 - 2a^n + q^n \geq 0$. Therefore

$$0 \leq \tau_n(a) \leq \frac{2}{1 - a^{2n}} - \frac{1}{1 - a^n} = \frac{1}{1 + a^n} \leq 1. \quad (5.5)$$

For $a > 1$, the two finite fixed points $\pm \sqrt{b/(a-1)}$ both have multiplier q . Formula (4.2) shows that their full primitive factor is precisely F_a from (1.6). Removing them means subtracting their contributions at *every* repetition:

$$\tau_n^{\text{red}}(a) = \frac{1}{a^n - 1} - \frac{2}{q^n - 1}, \quad D_a^{\text{red}}(u) = \exp \left(- \sum_{n \geq 1} \frac{\tau_n^{\text{red}}(a)}{n} u^n \right). \quad (5.6)$$

The factor F_a is nonzero on $|u| < 1$, so the quotient is unambiguous. Using (5.4) with $a = 1 + \varepsilon$, $q = 1 + 2\varepsilon$ proves $\tau_n^{\text{red}}(a) \rightarrow (n-1)/(2n)$. Convexity gives $q^n - 2a^n + 1 \geq 0$, and $q \leq a^2$ yields the uniform bound

$$0 \leq \tau_n^{\text{red}}(a) \leq \frac{1}{a^n - 1} - \frac{2}{a^{2n} - 1} = \frac{1}{1 + a^n} \leq \frac{1}{2}. \quad (5.7)$$

Fix $\rho < 1$. The summands of both normalized logarithmic series are bounded on $|u| \leq \rho$ by ρ^n/n , a summable sequence independent of the approaching parameter. Fixed-coefficient convergence and a finite-head, uniform-tail argument give uniform convergence of the logarithms. Exponentiation preserves it on this compact disk, proving both limits in Theorem 1.2.

The uncompensated limit is different. For real $0 < u < 1$,

$$\log D_a(u) \leq -\tau_1(a)u = -\frac{u}{a-1} \rightarrow -\infty \quad (a \downarrow 1), \quad (5.8)$$

whereas $D_a(0) = 1$. It cannot converge locally uniformly to D_1 , which is nonzero on the open disk. The required compensation is fixed by the two real orbits colliding with infinity, not by matching the desired limit after an arbitrary scalar normalization.

6 Circle ownership, operator limits and conclusion

For $a < 1$, infinity is a repelling fixed point of the projective-real map with local multiplier a^{-1} . Its primitive stability factor is

$$D_{\infty,a}(u) = \prod_{k \geq 1} (1 - a^k u). \quad (6.1)$$

Thus the full-circle product is

$$D_{\text{circle},a}(u) = D_a(u)D_{\infty,a}(u) = (1-u) \prod_{j \geq 1} (1 - q^j u)^2. \quad (6.2)$$

This is the classical Blaschke determinant mechanism [1]. Dividing out the specified orbit factor explains the poles in the finite-real product. However, deletion of a measure-zero orbit leaves ordinary L^p equivalence classes unchanged. Equation (6.2) is a periodic-product identity, not a theorem that restricting the same operator changes its trace by that factor. A natural physical transfer domain and a trace formula for it remain unspecified.

The Cayley scale $c = \sqrt{b/(1-a)}$ diverges at $a = 1$. Consequently the critical real map is not obtained by substituting $q = 1$ in a nonsingular fixed coordinate chart. The residue index and the all- n bounds provide the direct passage needed at the transition. They also show why matching finitely many coefficients would not establish either locally uniform limit.

The source contract is now closed: every iterate weight is explicit, every subcritical divisor coincidence is classified, and the critical germ has the stated two-sided compensated limit. Its classical unweighted phase structure was used to verify the physical periodic set, not counted as a separate contribution. The exceptional diagonal realizations are abstract source realizations only. No target primes, target zero correspondence, Euler factors, root numbers, automorphy, functional equation, or Hilbert–Pólya construction follows from these products.

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